

Ministry of Higher Education

Soran University

Software Engineering | Part 1 - Project Proposal

Second-hand Item Exchange System

GROUP MEMBERS

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SUPERVISOR

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1. Project Title

Second-hand Item Exchange System - a web-based platform that lets university students and local community members list, search for, exchange, and buy or sell used items safely and easily.

2. Problem Description

Many people own used items they no longer need, such as textbooks, electronics, furniture, clothes, and household goods. At the same time, others (especially students on a limited budget) need exactly these items. Today, the two sides are connected only through scattered social media groups, chat messages, and word of mouth.

This creates several problems: items are hard to find because posts get buried; there is no organized categorization or search; buyers cannot easily verify who they are dealing with; and there is no record of requests or exchanges, which leads to misunderstandings, scams, and wasted time. Perfectly usable items are thrown away or left unused, and students pay full price for things they could obtain second-hand.

The problem affects students, families, and small local communities. It matters because it has a direct financial effect, wastes useful resources, and can be solved effectively with a well-designed software system.

3. Target Users

- **Students** - main users who list unused items and look for affordable ones (books, laptops, furniture).
- **General community members** - people who want to sell, give away, or swap used household items.
- **Administrator** - manages users, categories, and listings, removes inappropriate content, and handles reports.

4. Project Objectives

- Provide a single organized platform where users can post second-hand items with photos, descriptions, and prices.
- Make finding items fast through categories, keyword search, and filters (price, condition, location).
- Allow safe communication and exchange or purchase requests between users, with request status tracking.
- Build trust through user accounts, ratings/reviews, and admin moderation of listings and reports.
- Encourage reuse of items and reduce the cost of everyday needs for students.

5. Proposed Solution

We propose a web application with a database backend. A registered user creates an account, posts an item with its details and images, and publishes it under a category. Other users can browse or search listings, view item details, and send an exchange or buy request to the owner. The owner can accept or reject requests, and once an item is exchanged it is marked as completed. After a transaction, users can rate each other.

An administrator dashboard allows the admin to approve or remove listings, block abusive accounts, and manage categories. The system will be designed using software engineering practices (requirements analysis, UML modelling, design, implementation, and testing) across the following project parts.

6. Project Scope

Included (In Scope)	Not Included (Out of Scope)
• User registration, login, and profile management • Add, edit, and delete item listings with images • Categories, search, and filtering • Exchange / buy request and status tracking • Messaging between buyer and owner • Ratings and reviews • Admin panel (users, listings, reports)	• Online payment gateway integration • Delivery and shipping services • Native mobile application • AI-based price or image recognition • Multi-language support beyond the basics

The scope is limited to a core web system so that the group can complete design, implementation, and testing within the remaining weeks of the semester.

7. Expected Benefits

- **Saves money** - students get needed items at low cost or by exchange.
- **Saves time** - one searchable place replaces scattered social media posts.
- **Reduces errors and scams** - verified accounts, ratings, and moderation increase trust.
- **Improves access to information** - clear item details, categories, and request history.
- **Supports sustainability** - reusing items reduces waste and builds a cooperative community.

8. Group Members and Responsibilities

No.	Student Name	Initial Responsibility
1	Lizaa Chato	Project coordination, requirements gathering, and documentation (final report compilation).
2	Bushra Faris	System analysis and UML modelling (use case, class, and sequence diagrams).
3	Passar Khdir	Database design and back-end development (users, items, requests).
4	Ariyan Walid	User interface design, front-end development, and testing.

Supervisor: Dr. Sirwan